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Adjuvant treatment with crude rhubarb for patients with acute organophosphorus pesticide poisoning: A meta-analysis of randomized controlled trials

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ABSTRACT

Background: Crude rhubarb has been used to treat critically ill patients for many years. However, no previous meta-analysis has been investigated the benefits of crude rhubarb in patients with acute organophosphorus pesticide poisoning (AOPP).

Objective: To summarize the beneficial effects of adjuvant treatment with crude rhubarb in patients with AOPP by conducting a meta-analysis.

Methods: A literature search of the databases through Pubmed, EMBASE, China National Knowledge Infrastructure, VIP, and Wanfang were performed for studies published up to October 2014. Randomized controlled trials (RCTs) investigating the effects of crude rhubarb as adjuvant treatment for patients with AOPP were included.

Results: A total of 12 RCTs with 886 patients were identified. Adjuvant treatment with crude rhubarb was associated with a significantly lower incidence of intermediate syndrome (risk ratio [RR] 0.22; 95% confidence interval [CI] 0.10–0.48), as well as multiple organ dysfunction syndrome (RR 0.34; 95% CI 0.20–0.56). Crude rhubarb as adjuvant treatment reduced the total dose of pralidoxime (mean difference [MD] −5.12 g; 95% CI −8.24 to −2.00) or atropine (MD −94.89 mg; 95% CI −156.22 to −33.57), and hospital length of stay (MD −2.79 days; 95% CI −4.19 to −1.39) compared with the controls.

Conclusions: This meta-analysis suggests that crude rhubarb as adjuvant treatment appears to have additional beneficial effects in patients with AOPP. More well-designed trials are needed to confirm our findings due to the methodological flaws of the included trials.

Contents

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1. Introduction

Self-poisoning with organophosphorus pesticides is a growing health problem. Acute organophosphorus pesticide poisoning (AOPP) causes approximately at least 233,997 deaths worldwide each year, accounting for 30% of suicides globally.[1] Organophosphorus is through the inhibition of acetylcholinesterase, resulting in the accumulation of acetylcholine at cholinergic synapses. Clinical manifestations of AOPP include acute cholinergic crisis, intermediate neurotoxic syndrome and delayed polyneuropathy. Current therapeutic strategies for patients with AOPP are mainly consisted of resuscitation, gastric lavage, intravenous atropine and pralidoxime, and supportive cares. However, these standard therapies were proving partly effective and the mortality rate was still up to 43.94% in Southeast Asia and India.[2] Therefore, novel therapies for the better management of AOPP are urgent.

Traditional Chinese Medicine (TCM) has been used for the management of AOPP for decades. Rhubarb, a species from polygonaceae family in the genus Rheum, is widely used Chinese herb. Rhubarb has been suggested to exert antioxidant activity,[3] antiplatelet aggregation activity,[4] and anti-inflammatory property.[5] There is a substantial body of published studies on crude rhubarb for the management of AOPP. In these studies, crude rhubarb is usually added to the standard therapies in AOPP patients. In addition to crude rhubarb alone, rhubarb-based formula, such as Qingxie Jiedu liquid,[6] Jiawei Xiaochengqi decoction,[7] Dahuang fuzi decoction[8] were also exerted promising results in the management of AOPP.

At present, no previously published meta-analysis has investigated the benefits of crude rhubarb as adjuvant treatment for patients with AOPP. Therefore, we conducted this meta-analysis to quantitatively summarize the add-on effect of crude rhubarb in patients with AOPP based on the available randomized controlled trials (RCTs).

2. Methods

2.1. Literature search

An extensive literature search of the databases through Pubmed, EMBASE, China National Knowledge Infrastructure, VIP, and Wanfang were performed for studies published up to October 2014. The pre-defined search terms included *Rheum palmatum L* OR rhubarb AND organophosphorus OR organophosphate OR organic phosphate OR poisoning AND random OR randomized controlled trial OR RCT. No restrictions in language were implemented. The reference lists of included studies were also manually screened to identify the additional trials.

2.2. Inclusion and exclusion criteria

Trails satisfied the following criteria were included: (1) RCTs investigating the benefits of crude rhubarb as adjuvant treatment for patients with AOPP; (2) AOPP was diagnosed on the history of exposure of a known organophosphorus pesticides, clinical features of poisonings, and decreased cholinesterase level; and (3) reporting at least one of the following outcomes: incidence of intermediate syndrome, death or multiple organ dysfunction syndrome (MODS), plasma cholinesterase level, total doses of pralidoxime or atropine, length of hospital stay or restoration time of cholinesterase. Intermediate syndrome is defined as the onset of progressive weakness after 24–96 h when acute cholinergic symptoms have resolved. Trials were excluded if: (1) the use of a formula containing rhubarb as intervention; (2) patients had received medical care in another medical center; and (3) patients had concomitant exposure of other herbal drugs.

2.3. Data extraction and quality assessment

Two reviewers independently extracted the following information from the included trials: first author's name, publication year, sample size, age range or mean age of patients, type of poisoning or poisoning periods, intervention, and main outcome measures. Cochrane risk of bias tool of RCTs was used to assess the methodological quality of the included trials. The trial's quality was judged as high, low, or unclear risk of bias according to the number of the meeting above criteria.

2.4. Statistical analysis

Comparisons of the effect sizes were made as crude rhubarb in combination with the usual treatment versus usual treatment alone. The outcomes were described in terms of mean difference (MD) or standardized mean difference (SMD) with 95% confidence interval (CI) for continuous data and the risk ratio (RR) with 95% CI for dichotomous data. Statistical heterogeneity was tested using the Cochrane Q statistic and I^2 index.[9] A fixed-effect model was chosen according to the heterogeneity (p value of Cochrane Q statistic test >0.10 and I^2 <50%); otherwise, a random effect model was selected. Publication bias was assessed by the visual inspection of the funnel plot. All the analyses were performed by the Cochrane Review Manager software (version 5.1, the Cochrane's collaboration, Oxford, UK). P value < 0.05 was considered as statistically significant.

3. Results

3.1. Search results

Our initial search identified 235 articles. Of which, 193 were excluded after screening the title and abstract. After retrieving and analyzing the full text papers, 30 papers were further excluded. Finally, a total of 12 trials[10–21] were eligible for the meta-analysis. No additional trials were found by manually searching reference list of the relevant studies. The detail of trials selection process is presented in Fig. 1.

3.2. Trials characteristics

The characteristics of the included trials are described in Table 1. The total number of patients included in the present meta-analysis was 886 (448 in the crude rhubarb group, 438 in the control group). All of the trials were published from 2004 to 2014 and performed in China. The dosage of crude rhubarb ranged from 9 to 120 g per day. Crude rhubarb was administered by gastric tube, nasal feed-

Table 1
Characteristics of the included trials in the meta-analysis.

Study/year	No. patients	Age (years)	Type of poisoning (number)	Poisoning periods (h)	Main intervention		Outcome measures
	Rhubarb/control	Rhubarb/control			Rhubarb group	Control (UT) group	
Zhang et al. [10]	31/31	32.1 ± 16.1	Omethoate (15), thimet (16), parathion (10), dichlorvos (14), monocrotophs (7)	0.5–6.2	Rhubarb 60g/d × 3d (GT) + UT	Atropine, pralidoxime, support and symptomatic therapy	①+②
Yang et al [11]	44/43	41.93 ± 17.63/41.38 ± 19.76	NP	NP	Rhubarb 60g/d × 7d (GT) + UT	Gastrolavage, catharsis, respiratory support, dehydration, diuresis, atropine, and pralidoxime.	⑦
Zhang et al. [12]	22/21	34.3 ± 16.7/37.2 ± 14.3	Methamidophos (11), parathion (9), dichlorvos (19), others (4)	1.8 ± 1.20/ 1.4 ± 0.9	Rhubarb 15–30 g/d (NF) + 15–30 g (RE) + UT	Atropine, pralidoxime, anti-infection, gastrolavage, catharsis, support and symptomatic therapy	①
Zhou et al. [13]	59/58	NP	Dichlorvos (117)	0.27-6	Rhubarb 40–120 g/d (GT) + UT	Atropine, pralidoxime, oxygen inhalation, tracheal intubation, anti-infection, gastrolavage, and catharsis	①+②
Tan et al. [14]	30/28	30.5 ± 12.2/31.0 ± 12.7	Methamidophos (20), omethoate (15), dichlorvos (11), parathion (12).	NP	Rhubarb 30 g/d (OA or GT) + UT	Atropine, pralidoxime, tracheal intubation, gastrolavage, mechanical ventilation, support and symptomatic therapy	②
Ren et al. [15]	23/25	36.09 ± 10.64/36.09 ± 18.64	Methamidophos (19), omethoate (13), dichlorvos (9), dimethoate (3), phoxim (4)	1.26 ± 0.86/ 1.26 ± 0.74	Rhubarb 15–30 g/d (GT) + UT	Atropine, pralidoxime, gastrolavage, catharsis, mechanical ventilation, diuretic, support and symptomatic therapy	①
Yang et al. [16]	35/33	18–83	Dichlorvos (25), omethoate (15), dimethoate (15), trichlorfon (13)	0.5-6	Rhubarb 2–10 g/d × 8d (GT) + UT	Gastrolavage, catharsis, mechanical ventilation, diuresis, atropine, pralidoxime, support and symptomatic therapy	②+③+④+⑤+⑥
Xue et al. [17]	45/44	16–56	Dichlorvos (9), omethoate (6), parathion (34), methamidophos (32), dimethoate (8),	NP	Rhubarb 9 g/d × 7d (OA or NF) + UT	Atropine, pralidoxime, and other support therapies	③
Hou [18]	50/52	38.53 ± 9.11/36.27 ± 10.15	Dichlorvos (28), methamidophos (41), dimethoate (37), malathion (22)	3.50 ± 1.20/ 4.10 ± 1.00	Rhubarb 12 g/d × 7d (OA or GT) + UT	Atropine, pralidoxime, gastrolavage,cleaning the skin, activated carbon, support and symptomatic therapy	①+④+⑤+⑥+⑦
Chen et al. [19]	31/31	16–71	Dimethoate (21), methamidophos (30), dichlorvos (11).	NP	Rhubarb 40 g/d (NF) + UT	Atropine, pralidoxime, gastrolavage, diuresis, mechanical ventilation, keep electrolyte and acid-base balance.	①+⑦
Chen et al. [20]	36/29	14–51	Dichlorvos (18), dimethoate (18), trichlorfon (11), methamidophos (18)	NP	Rhubarb 10–20 g/d (NF) + 20–40 g (RE) × 3–5d + UT	Atropine, pralidoxime, anti-infection, tracheal intubation, gastrolavage, diuresis, antishock, and support therapy.	②+③+⑦
Wang et al. [21]	42/43	51.6 ± 13.5/53.2 ± 14.1	Dichlorvos (29), parathion (21), dimethoate (20), trichlorfon (7),others (8)	NP	Rhubarb 10–20 g/d (NF) + UT	Atropine, pralidoxime, gastrolavage, catharsis, support and symptomatic therapy,keep electrolyte and acid-base balance.	⑥+⑦

Abbreviations: MODS, multiple organ dysfunction syndrome; NF, nasal feeding; RE, retention enema; GT, gastric tube; NP, not provide; UT, usual treatment. ① Intermediate syndrome; ② mortality rate; ③ MODS; ④ levels of cholinesterase; ⑤ total dose of pralidoxime; ⑥ total dosage of atropine; ⑦ hospital stay.混悬液中大黄含量–馆 使患者每日排便次

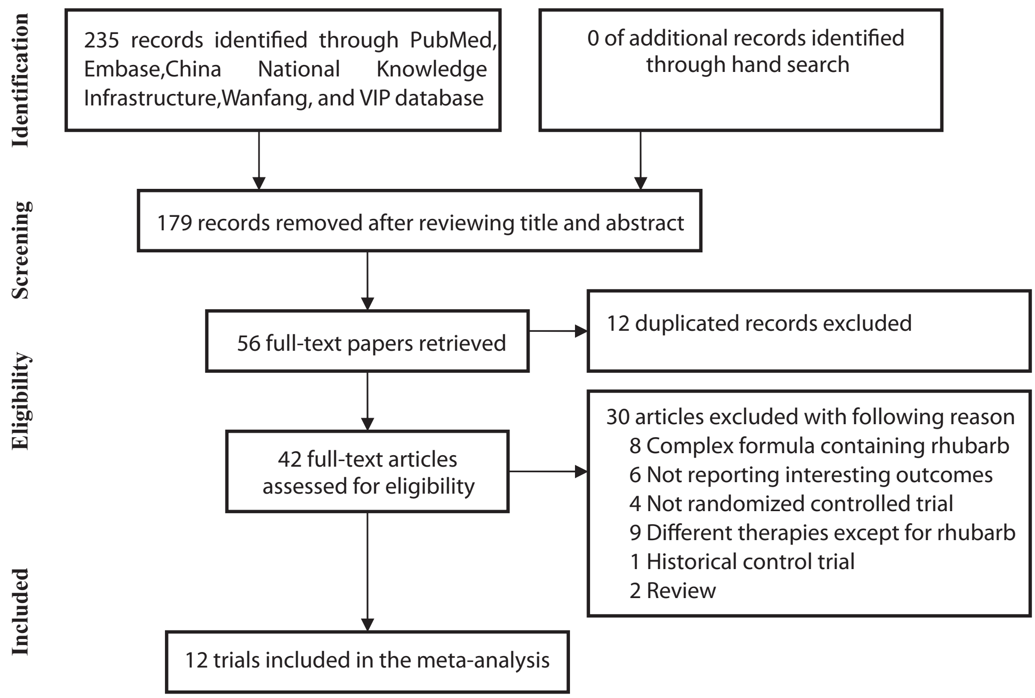

Fig. 1. Flow chart showing the trials selection process.

ing, and retention enema. Quality assessment indicated that lack of sufficient information to judge the detailed method of randomization and allocation concealment was the main sources of potential risk bias. In addition, the domains of blinding outcome assessment, incomplete data reporting, and selective outcome reporting were grouped as low risk of bias except for two study[11,17] (Supplemental Fig. S1).

3.3. Effects of rhubarb on cholinesterase level and total dose of pralidoxime or atropine

Two included trials[16,18] reported changes in plasma cholinesterase level. As it is shown in Fig. 2A, a random effect model was selected for the analysis because of the significant heterogeneity between the 2 trials ($I^2 = 98\%$, $p < 0.001$). There was no statistically significance in serum cholinesterase level (SMD 2.40; 95% CI −1.06 to 5.85) between two groups. As it is shown in Fig. 2B, two trials[16,18] reported the consumption of total dose of pralidoxime. A random effect model was chosen for the analysis because of the obvious heterogeneity between the 2 trials ($I^2 = 56\%$, $p = 0.13$). The pooled estimate showed a significant reduction in total dose of pralidoxime (MD −5.12 g; 95% CI −8.24 to −2.00) in crude rhubarb plus usual treatment group compared with the usual group. As it is shown in Fig. 2C, 4 trials[11,16,18,21] reported the total dose of atropine. As for there was a significant amount of heterogeneity across the trials ($I^2 = 93\%$, $p < 0.001$), a random effect model was selected for the analysis. The pooled estimate showed adjuvant treatment with crude rhubarb reduced the total dose of atropine (MD −94.89 mg; 95% CI −156.22 to −33.57) compared with the usual treatment group.

3.4. Effects of crude rhubarb on clinical outcomes

Six trials[10,12,13,15,18,19] reported the incidence of intermediate syndrome. As it is shown in Fig. 3A, a fixed-effect model was selected for the analysis because there was no evidence of significant heterogeneity ($I^2 = 0\%$, $p = 0.98$) across the trials. The pooled estimate indicated a significant reduction in intermediate syndrome rate (RR = 0.22; 95% CI 0.10–0.48) in crude rhubarb plus usual treatment group compared with the usual group. As it is shown in Fig. 3B, the pooled estimate from 4 trials[10,16,17,20] showed adjuvant treatment with crude rhubarb reduced MODS rate (RR 0.34; 95% CI 0.20–0.56) compared with the usual treatment group in a fixed-effect model. Five trials[10,13,14,16,20] reported outcomes of mortality rate. As it is shown in Fig. 3C, the mortality rate in the crude rhubarb and usual treatment group was 10 out of 191 deaths (5.24%) and 17 out of 179 deaths (9.50%). The pooled estimate indicated some trend in decreasing mortality rate (RR 0.52; 95% CI 0.25–1.08), in which this reduction was not statistically significant ($p = 0.08$). Four trials[18–21] provided the outcome of length of hospital stay. As it is shown in Fig. 4, there was a significant amount of heterogeneity ($I^2 = 84\%$, $p < 0.001$) across the trials, so we selected the random effect model. The pooled estimate indicated adjuvant treatment with crude rhubarb reduced the length of hospital stay (MD −2.79 days; 95% CI 4.19 to −1.39) compared with the usual treatment group.

3.5. Subgroup analyses and publication bias

We conducted subgroup analyses on the incidence of intermediate syndrome and consumption of total dose of atropine which included a larger number of trials. Findings from subgroup analyses indicated that the dose of crude rhubarb over 30 g/day achieved a greater reduction in intermediate syndrome rate (Supplemental Fig. S2) and total dose of atropine (Supplemental Fig. S3). Visual inspection of funnel plots of intermediate syndrome rate or mortality rate were symmetrical, suggesting no evidences of publication bias (Supplemental Fig. S4).

3.6. Crude rhubarb versus mannitol intervention

Moreover, we conducted an additional meta-analysis on crude rhubarb versus mannitol as adjuvant treatment in patients with AOPP with the available RCTs. Four trials[22–25] were eligible for the analysis. Characteristics of included trials are shown in Table 2. As it shown in Fig. 5, compared with the mannitol treatment, adju-

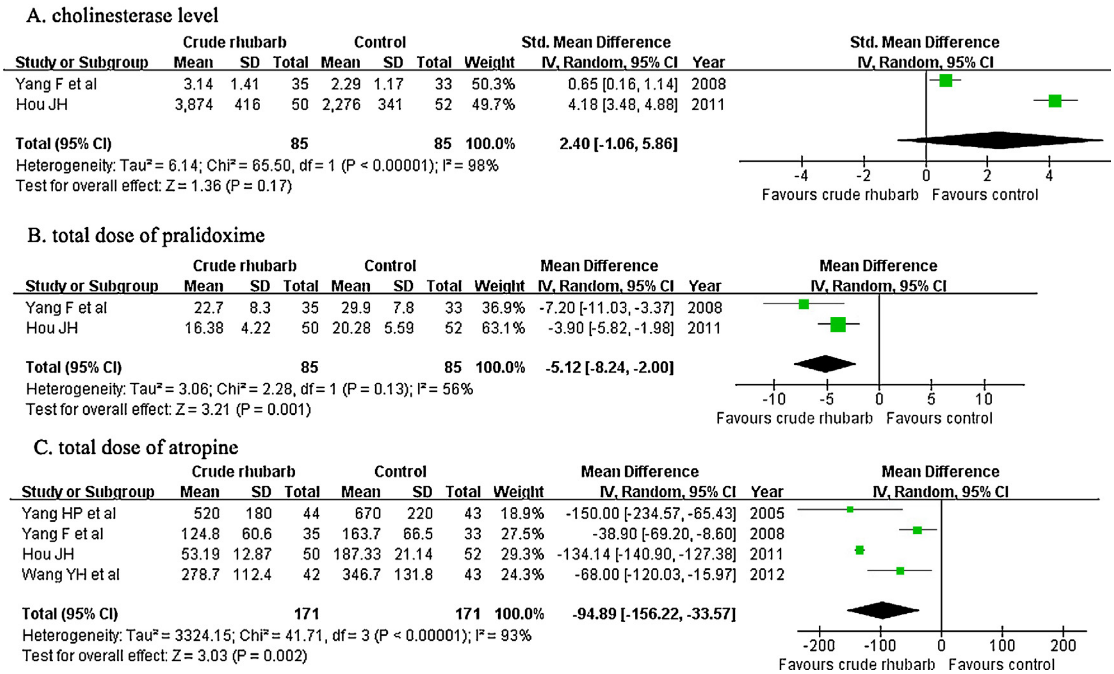

Fig. 2. Forest plots showing comparison of plasma cholinesterase level (A), total dose of pralidoxime (B) or atropine (C) comparing crude rhubarb to the usual treatment.

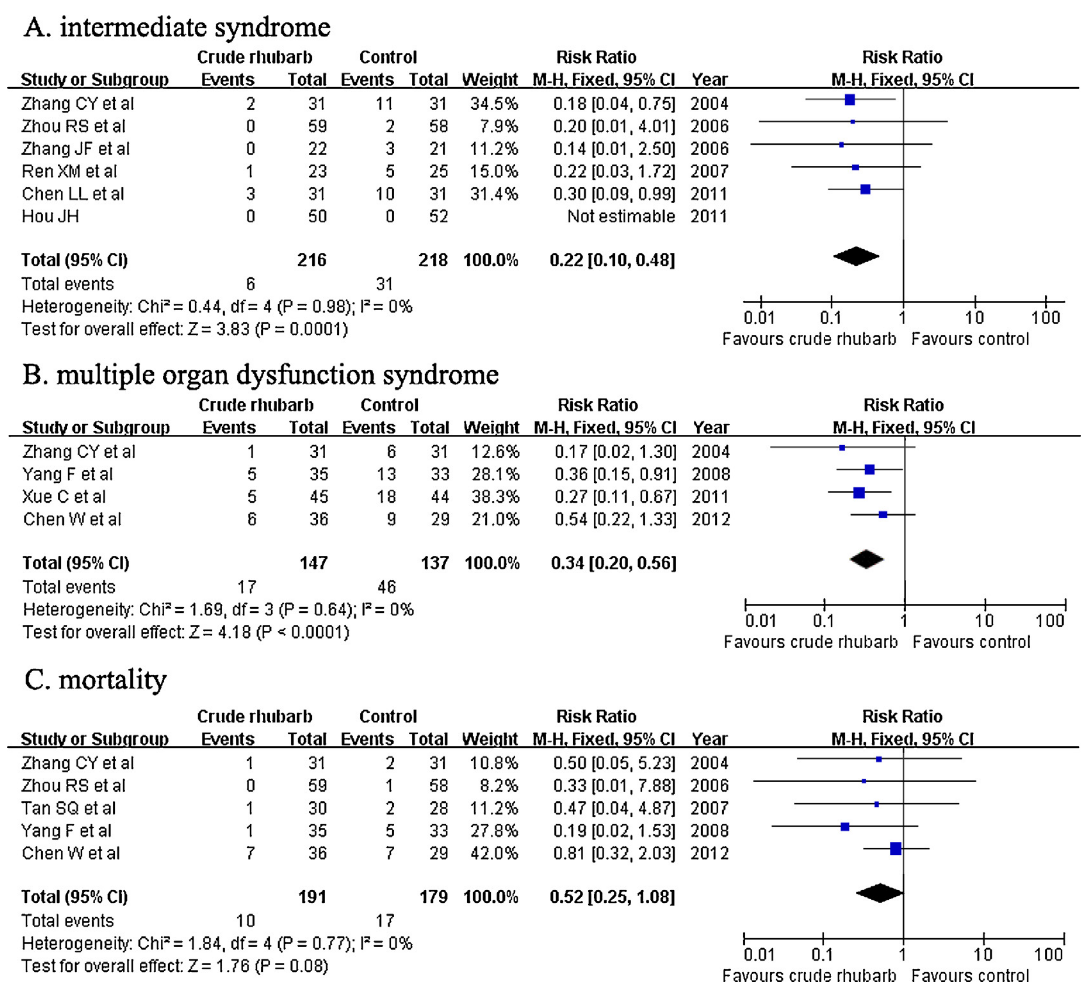

Fig. 3. Forest plots showing comparison of incidence of intermediate syndrome (A), multiple organ dysfunction syndrome (B), and mortality (C) comparing crude rhubarb to the usual treatment.

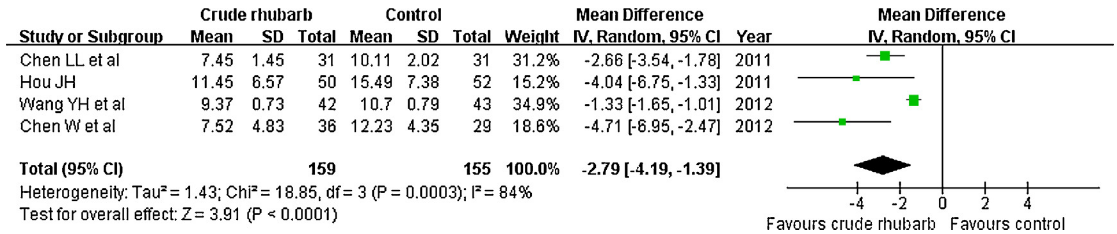

Fig. 4. Forest plots showing comparison of length of hospital stay comparing crude rhubarb to the usual treatment.

Table 2
Characteristics of the included trials in the additional analysis.

Study/year	No. patients	Age (years)	Type of poisoning (number)	Duration of poisoning (h)	Intervention		Outcome measures
	Rhubarb/mannitol	Rhubarb/mannitol			Rhubarb group	Mannitol group	
Li [22]	36/32	16–56	Dichlorvos(9), dimethoate (41), parathion (10), methamidophos (4), dipterex (2), demeton (1)	0.3–0.6	Crude rhubarb powders 20 g/d (OA or NF) + UT	20% Mannitol 250 ml/d (GT) + UT	①+②+③
Kong [23]	32/32	17–67	NP	NP	Crude rhubarb powders 20–30 g/d (GT) + UT	20% Mannitol 250 ml/d (GT) + UT	②
Xia et al. [24]	20/20	NP	Omethoate plus dimethoate (18), dipterex (8), dichlorvos (9), others (5)	NP	Crude rhubarb powders 10-30 g/d(GT)+ UT	20% Mannitol 250 ml/d (GT) + UT	②+③
Liu [25]	40/36	NP	Methamidophos (22), dimethoate (28), dichlorvos (20), malathion (6).	0.27–6	Crude rhubarb powders 20 g/d (GT) + 13.3 g (RE)/d + UT	20% Mannitol 150 ml/d (GT) + UT	① +②+③

Abbreviations: NF, nasal feeding; RE, retention enema; GT, gastric tube; NP, not provide; UT, usual treatment. ① Total dosage of atropine; ② hospital stay; ③ Restoration time of cholinesterase. *Usual treatment includes atropinization, pralidoxime chloride orpralidoxime, and other support therapies.

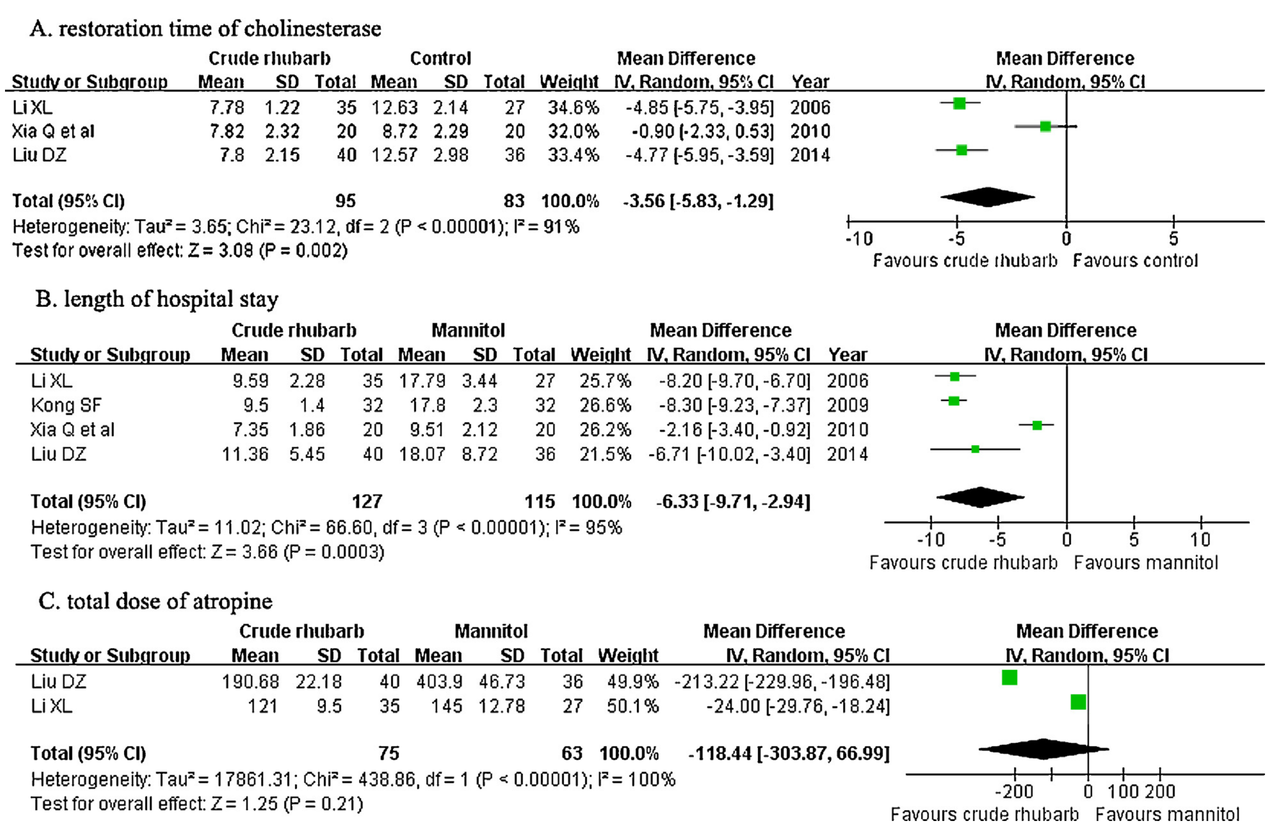

Fig. 5. Forest plots showing comparison of restoration time of cholinesterase (A), length of hospital stay (B), and total dose of atropine (C) comparing crude rhubarb to mannitol as adjuvant treatment.

vant treatment with crude rhubarb reduced the restoration time of cholinesterase (MD −3.56 days; 95% CI −5.83 to −1.29) and length of hospital stay (MD −6.33 days; 95% CI −9.71 to −2.94). However, there was not any statistical significance in the total dose of atropine (MD −118.44 mg; 95% CI −303.87 to 66.99).

4. Discussion

To the best of our knowledge, this is the first meta-analysis investigating the benefits of crude rhubarb in patients with AOPP. Crude rhubarb as adjuvant treatment appears to reduce the incidence of intermediate syndrome, mortality or MODS, total dose of pralidoxime or atropine in patients with AOPP. In addition, crude rhubarb was superior to mannitol as adjuvant treatment in restoration time of cholinesterase level and length of hospital stay among AOPP patients.

Estimation of blood cholinesterase activity is a screening biomarker for organophosphorus poisoning. Reverse inhibition of acetylcholinesterase by treatment with pralidoxime might confer some benefits.[26] Atropine is the universal treatment for organophosphorus anticholinesterase poisoning. Despite atropine and pralidoxime are traditionally applied in the management of organophosphorus poisoning, their efficacy remains a major issue of debate. In this study, adjuvant treatment with crude rhubarb reduced the total dose of pralidoxime and atropine compared with the usual treatment in patients with AOPP. Results from subgroup analysis indicated that the dose of crude rhubarb over 30 g/day achieved a greater reduction in total dose of atropine. These findings suggested crude rhubarb as adjuvant treatment had additional benefits in the management of AOPP.

In addition to above mentioned benefits, crude rhubarb also reduced the organophosphate-related complications. Intermediate neurotoxic syndrome typically occurs within 24–96 h and characterizes by cranial nerve palsies, weakness of the neck and proximal limbs, and respiratory paralysis. Intermediate neurotoxic syndrome was the main cause of organophosphate-related morbidity and mortality because of its consequence of respiratory failure.[27] Unfortunately, there are no specific antidotes available for this syndrome. Therefore, it is important to explore the new therapies. Compared with the usual treatment alone, adjuvant treatment with crude rhubarb had additional benefits in reducing the incidence of intermediate syndrome. Moreover, there was fewer death rates (5.24% vs. 9.50%) in patients who had received crude rhubarb compared with the controls. In addition, patients who received crude rhubarb treatment reduced 2.79 days hospital stay than those who have not received crude rhubarb.

Rhubarb mainly contains anthraquinones, anthrones, flavonoids, acylglucosides, stilbenes, and pyrones. In TCM, rhubarb is usually used as a laxative to unclog the bloodstream, eliminate internal toxins, and purgation. Subgroup analysis indicated that the dose of crude rhubarb over 30 g/day appears to achieve better clinical effects than the low-dose rhubarb. However, it should be noted that current dosage of crude rhubarb was more than the restricted dose in the Chinese Pharmacopoeia. Eliminating internal toxins and purgation is the main efficacies of rhubarb. So mild diarrhea is a normal phenomenon of crude rhubarb used. The appropriate dosage of crude rhubarb was adjusted according to the stool frequency (up to 4 times per day) in most studies. Despite not any studies reported high-dose rhubarb-related electrolyte imbalance events, the potential adverse effects of high-dose of rhubarb need to be closely monitored.

Management of constipation is the most frequent application of rhubarb owing to its purgative property.Whether the benefits of crude rhubarb are determined by its purgative property is unclear. Use of mannitol for gastric lavage resulted in diarrhea. Our additional analysis based on the crude rhubarb versus mannitol as adjuvant treatment showed that crude rhubarb was superior to mannitol in the restoration time of cholinesterase and reduction in hospital stay. These findings indicated that the purgative property was not the sole mechanism of crude rhubarb. So the possible mechanisms of crude rhubarb are needed to be further explored.

This meta-analysis has several limitations. First, despite all the included trials claimed the randomized allocation of participants, but only two trials[15,20] described the detailed methods of randomization, and none of trials provided the allocation concealment, sample size calculation or the number of dropouts. These methodological weaknesses of the included studies limited the generalizability of the utilization of crude rhubarb in AOPP patients. Second, despite visual inspection of the funnel plots did not find evidence of publication bias; however, all the included studies were published in Chinese and lack of other language literatures; therefore, potential publication bias cannot be excluded. Third, although high-dose crude rhubarb appears to achieve greater benefits, we still cannot determine the appropriate regimen of crude rhubarb due to the individual trial used different doses of crude rhubarb and duration of intervention. Moreover, regimens of crude rhubarb and severity of AOPP patients might be partly explained the significant heterogeneity in pooling the continuous outcomes. Fourth, rhubarb-based traditional Chinese prescriptions are also widely used in AOPP patients. However, we did not include these formulas containing crude rhubarb in the meta-analysis because of significant heterogeneity in the herbal medicines. Finally, organophosphorus insecticide is associated with substantial variability in the clinical course and response to treatment.[26] Difference in organophosphorus insecticide exposure between groups might have affected our findings.

5. Conclusions

The results of this meta-analysis suggest that adjuvant treatment with crude rhubarb may offer additional benefits on reducing the incidence of intermediate syndrome and MODS, total dose of pralidoxime or atropine in patients with AOPP. However, evidence of rhubarb benefits patients with AOPP cannot be provided due to the methodological flaws of the included trials. Therefore, our findings need to be further confirmed by more well-designed RCTs. Moreover, the underlying mechanisms of rhubarb in the treatment of AOPP are needed to be further investigated.

Conflict of interests

None.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.ctim.2015.09.001.

References

1. Gunnell D, Eddleston M, Phillips MR, Konradsen F. The global distribution of fatal pesticide self-poisoning: systematic review. *BMC Public Health*. 2007;7:357.
2. Dash SK, Mohanty MK, Mohanty S, Patnaik KK. Organophosphorus poisoning: victim specific analysis of mortality and morbidity. *Med Sci Law*. 2008;48:241–245.

3. Matsuda H, Morikawa T, Toguchida I, Park JY, Harima S, Yoshikawa M. Antioxidant constituents from rhubarb: structural requirements of stilbenes for the activity and structures of two new anthraquinone glucosides. *Bioorg Med Chem*. 2001;9:41–50.
4. Harima S, Matsuda H, Kubo M. Study of various rhubarbs regarding the cathartic effect and endotoxin-induced disseminated intravascular coagulation. *Biol Pharm Bull*. 1994;17:1522–1525.
5. Peng SM, Wang SZ, Zhao JP. Effect of rhubarb on inflammatory cytokines and complements in patients with systemic inflammation reaction syndrome and its significance. *Zhongguo Zhong Xi Yi Jie He Za Zhi*. 2002;22:264–266.
6. Zuo JL, Zhou HP, Chen P. Clinical observation of Qingxie Jiedu liquid on 22 cases of organophosphorus poisoning patients. *J New Chin Med*. 2006;38:48–49.
7. Han ZB, Wang HW. Clinical observation of Jiawei Xiaochengqi Decoction on the oral organophosphorus pesticide poisoning. *Chin J Crit Care Med*. 2001;21:611–612.
8. Fang XL, Wang SM. Observation of the ciinical effect of treatment with warm and purgation method in acute organophosphorus pesticides poisoning. *JETCM*. 2014;23:1270–1271, 306.
9. Higgins JP, Thompson SG, Deeks JJ, Altman DG. Measuring inconsistency in meta-analyses. *BMJ*. 2003;327:557–560.
10. Zhang CY, Chi JF, Zhao YM, Yang CL, Feng JH, Chang ZQ, et al. Clinical value of rhubarb purgation in the treatment of acute organophosphorus pesticide poisoning. *Shandong Med J*. 2004;44:1–3.
11. Yang HP, Kang XG. Clinical observation of rhubarb on acute severe organophosphorus pesticide poisoning induced gastrointestinal function failure Shaanxi. *J Tradit Chin Med*. 2005;26:1185–1186.
12. Zhang JF, Guo R, Zhang JP, Liao W. Clinical observation of rhubarb in acute organophosphorus pesticide poisoning. *J Gansu College TCM*. 2006;23:45–46.
13. Zhou RS, Liu F. Clinical experience of rhubarb treatment for dichlorvos poisoning. *JETCM*. 2006;15:1163.
14. Tan SQ, Xiao GL, Wang ZY, Tang ZX, Zhang CY, Zhang L. Effect of rhubarb on C-reactive protein in patients with acute oral organophosphorus pesticide poisoning. *J TCM Univ Hunan*. 2007;27:60–64.
15. Ren XM, Liu FL. Effect of rhubarb on prevention and treatment for intermediate syndrome of acute organophosphorus pesticide poisoning. *J Med Forum*. 2007;28:115–116.
16. Yang F, Ke ZY, Kang JY. Effect of rhubarb on severely acute organophosphorus pesticide poisoning. *Chin J Med Guide*. 2008;10:887–888.
17. Xue C, Wang WH. Therapeutic and preventive effects of rhubarb on multiple organ dysfunction syndrome in patients with acute organophosphorus pesticide poisoning. *J Wannan Med College*. 2011;30:308–310.
18. Hou JH. Activated charcoal treatment for 50 cases of acute severe organophosphorous pesticide poisoning. *JETCM*. 2011;20:997–998.
19. Chen LL, Wu JL, Fu M, Wang LX, Zhang XH, Hao XF. Clinical experience of combined application of rhubarb in the treatment of acute severe organophosphorus pesticide poisoning. *China Prac Med*. 2011;6:168–169.
20. Chen W, Sun LJ. Effect of rhubarb on complications of acute severe organophosphorus pesticide poisoning and prognosis. *JETCM*. 2012;21:641.
21. Wang YH, Zhang HT. Clinical effect of adjuvant treatment with rhubarb on acute severe organophosphorus pesticide poisoning. *JETCM*. 2012;21:2045–2046.
22. Li XL. Clinical observation of rhubarb treatment for 35 cases of acute organic phosphorus poisoning. *J New Chin Med*. 2006;38:54–55.
23. Kong SF. Investigation of cathartic method on the organophosphorus pesticide poisoning. *J Nurses Train*. 2009;24:2196–2197.
24. Xia Q, Shi Y, Shi MM. Clinical analysis of penehyclidine hydrochloride combined with rhubarb in treatment of severe organophosphorus pesticide poisoning. *Chin J Misdiagn*. 2010;10:5124.
25. Liu DZ. Clinical effect of rhubarb as nasal feeding, retention enema in the treatment of acute organophosphorus poisoning and impact on gastrointestinal function. *JETCM*. 2014;23:981–982.
26. Eddleston M, Eyer P, Worek F, Mohamed F, Senarathna L, von Meyer L, et al. Differences between organophosphorus insecticides in human self-poisoning: a prospective cohort study. *Lancet*. 2005;366:1452–1459.
27. Yang CC, Deng JF. Intermediate syndrome following organophosphate insecticide poisoning. *J Chin Med Assoc*. 2007;70:467–472.